

- Lab 7 : Data Analytics
- Lecturer : Prof. Youness Moukafih
- Lab instructor: Prof. Berkani Safaa

Lab 7 - Probabilities and Distributions

Objective:

In this lab, you'll use Python to explore and visualize various probability concepts. Complete the following exercises, using the hints provided when needed.

Setup

Before starting, make sure you have the following libraries imported:

```
import numpy as np
import matplotlib.pyplot as plt
from scipy import stats
```

Set a random seed for reproducibility:

```
np.random.seed(35)
```

Exercise 1: Sampling and Law of Large Numbers

Create a simulation of coin flips to demonstrate the Law of Large Numbers.

- Write a function `coin_flip_experiment(num_flips)` that simulates flipping a coin `num_flips` times.
Hint: Use `np.random.choice()` with appropriate arguments.
- Create a function `plot_coin_flip_convergence(max_flips)` that plots the proportion of heads vs. the number of flips.
- Run your simulation with 10,000 flips and plot the results.

Questions:

- How does the proportion of heads change as the number of flips increases?
- What does this demonstrate about the Law of Large Numbers?
- Try running the simulation multiple times. Do you get the exact same result each time? Why or why not?

Exercise 2: Discrete Probability Distributions

In this exercise, you'll work with the Binomial distribution.

- Create a function `plot_binomial_distribution(n, p)` that plots the probability mass function of a Binomial distribution with parameters `n` (number of trials) and `p` (probability of success).
Hint: Use `stats.binom.pmf()` to calculate the probabilities.
- Plot the Binomial distribution for `n=20` and `p=0.5`.

Questions:

- What does this distribution represent in terms of a real-world scenario?
- How does the shape of the distribution change if you increase `n` to 50? What if you change `p` to 0.7?
- Can you think of a situation where this distribution might be useful?

Exercise 3: Expected Value and Variance

Simulate the card game described in the lecture where you win: \$1 if you draw a heart (not including the ace of hearts), \$5 if you draw an ace (including the ace of hearts), \$10 if you draw the king of spades, and \$0 for any other card.

- Create a function `card_game_simulation(num_games)` that simulates playing this game `num_games` times.
Hint: Use `np.random.choice()` with appropriate probabilities.
- Write a function `analyze_card_game(num_games)` that:
 - Runs the simulation.
 - Calculates and prints the mean, variance, and standard deviation.
 - Plots a histogram of the winnings.
- Run your analysis with 10,000 games.

Questions:

- What is the expected value of playing this game once? Is it a fair game?
- How does the standard deviation relate to the spread of outcomes you see in the histogram?
- If you were to play this game 100 times, would you expect to always win the average amount? Why or why not?

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Exercise 4: Continuous Probability Distributions

In this exercise, you'll work with the Normal distribution.

- a) Create a function `plot_normal_distribution(mean, std_dev)` that plots the probability density function of a Normal distribution.

Hint: Use `np.linspace` to generate values on the x-axis.

Use the `stats.norm.pdf()` to calculate the probabilities.

- b) Plot the Normal distribution with mean 170 and standard deviation 10.
- c) Shade the area within one standard deviation of the mean.

Questions:

1. What might this distribution represent in a real-world context?
2. Approximately what percentage of the data falls within one standard deviation of the mean?
3. How would the shape of the distribution change if you decreased the standard deviation to 5?

End of the lab.